

10P SHINE INTERNSHIP PROGRAM – COHORT 8

Assignment No. 4 – API Testing with Postman

Project Title: Restful API (<https://api.restful-api.dev>)

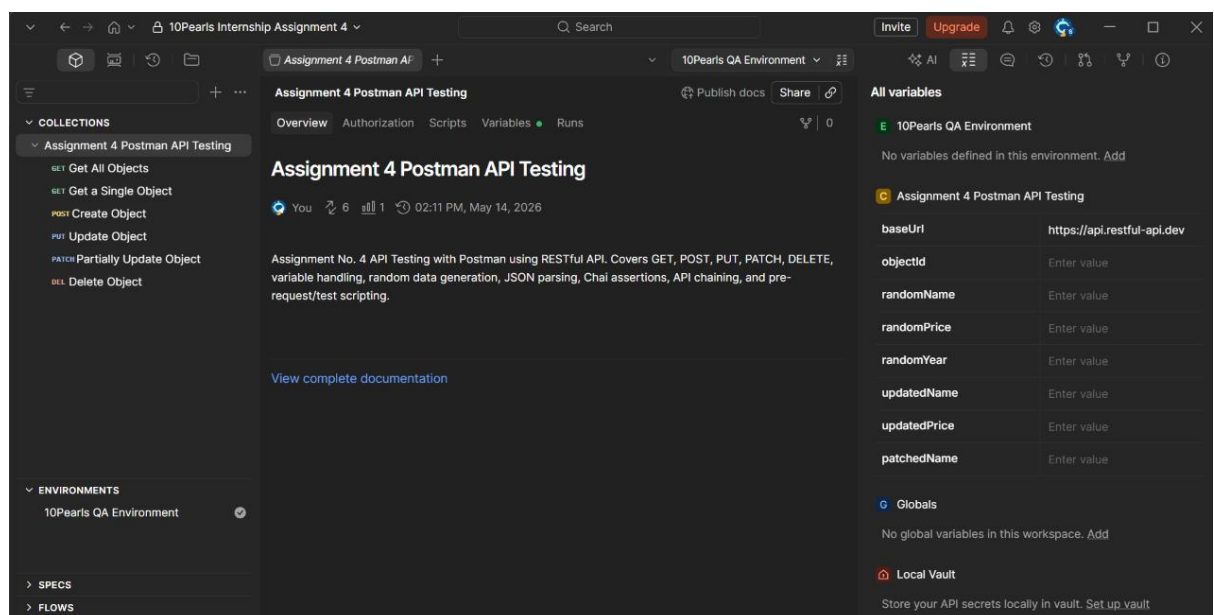
Date: May 14, 2026

Submitted By: Syeda Azwa Asif

1. PROJECT OVERVIEW

API Testing with Postman using RESTful API. Covers GET, POST, PUT, PATCH, DELETE, variable handling, random data generation, JSON parsing, Chai assertions, API chaining, and pre-request/test scripting.

2. CREATING COLLECTION



3. VARIABLES IN COLLECTION

The screenshot shows the Postman interface with the 'Variables' tab selected for the collection 'Assignment 4 Postman API Testing'. The interface includes a sidebar with a 'COLLECTIONS' list and an 'ENVIRONMENTS' list. The main area displays a table of variables.

Variable	Value
baseUrl	https://api.restful-api.dev
objectId	
randomName	
randomPrice	
randomYear	
updatedName	
updatedPrice	
patchedName	
Add variable	

4. GET ALL OBJECTS

The screenshot shows the Postman interface with the 'Scripts' tab selected for the request 'Get All Objects'. The interface includes a sidebar with a 'COLLECTIONS' list and an 'ENVIRONMENTS' list. The main area displays the script code for the request.

```
1 const objects = pm.response.json();
2 console.log('Total objects in the DB: ${objects.length}');
3
4 if (objects.length > 0) {
5   console.log('First object in list:', objects[0].name);
6   pm.test('First object has id and name', () => {
7     pm.expect(objects[0]).to.have.property('id');
8     pm.expect(objects[0]).to.have.property('name');
9   });
10 } else {
11   console.warn('No objects found - run Create Object first');
12 }
13
14 pm.test('Status code is 200', () => {
15   pm.response.to.have.status(200);
16 });
17
18 pm.test('Response is an array', () => {
19   pm.expect(objects).to.be.an('array');
20 });
```

5. GET A SINGLE OBJECT

The screenshot shows the Postman interface for a GET request named "Get a Single Object". The URL is `{{baseUrl}}/objects/{{objectId}}`. The "Scripts" tab is active, showing the following code:

```
3
4 console.log('Retrieved Single Object');
5 console.log('ID: ${obj.id}, Name: ${obj.name}');
6 console.log('Data:', obj.data);
7
8 pm.test('Status code is 200', () => {
9   pm.response.to.have.status(200);
10 });
11
12 pm.test('Retrieved ID matches chained objectId', () => {
13   pm.expect(obj.id).to.equal(expectedId);
14 });
15
16 pm.test('Object has name and data properties', () => {
17   pm.expect(obj).to.have.property('name');
18   pm.expect(obj).to.have.property('data');
19 });
```

The "Response" tab is selected, showing a successful response with status 200. The response body is not visible, but the status is 200.

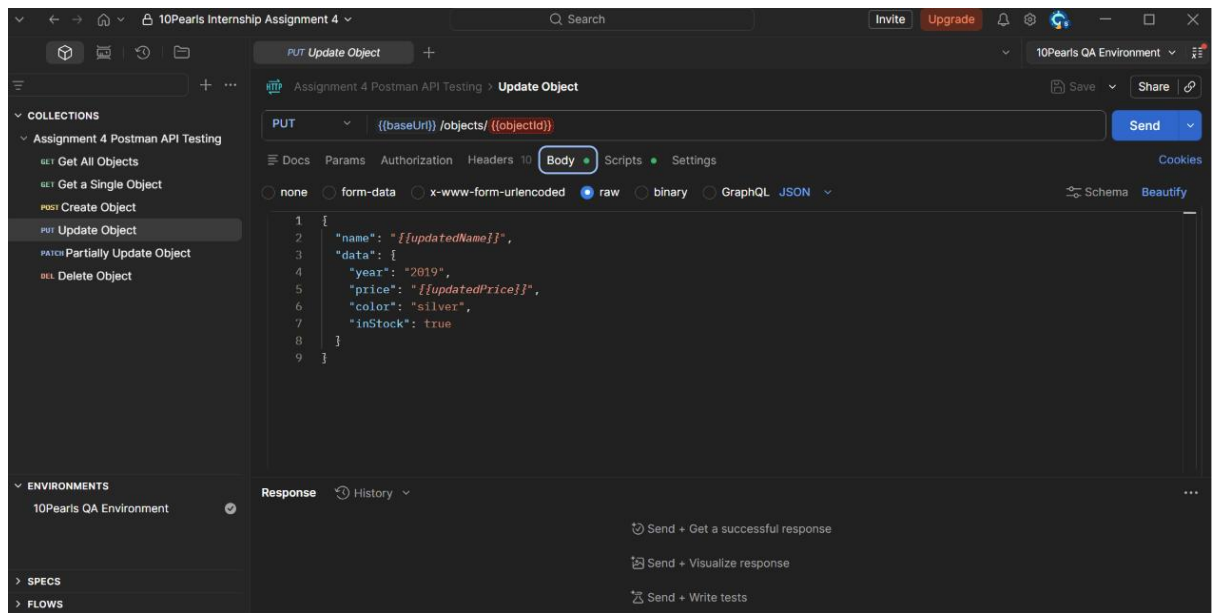
6. CREATE OBJECT

The screenshot shows the Postman interface for a POST request named "Create Object". The URL is `{{baseUrl}}/objects`. The "Body" tab is active, showing the following JSON body:

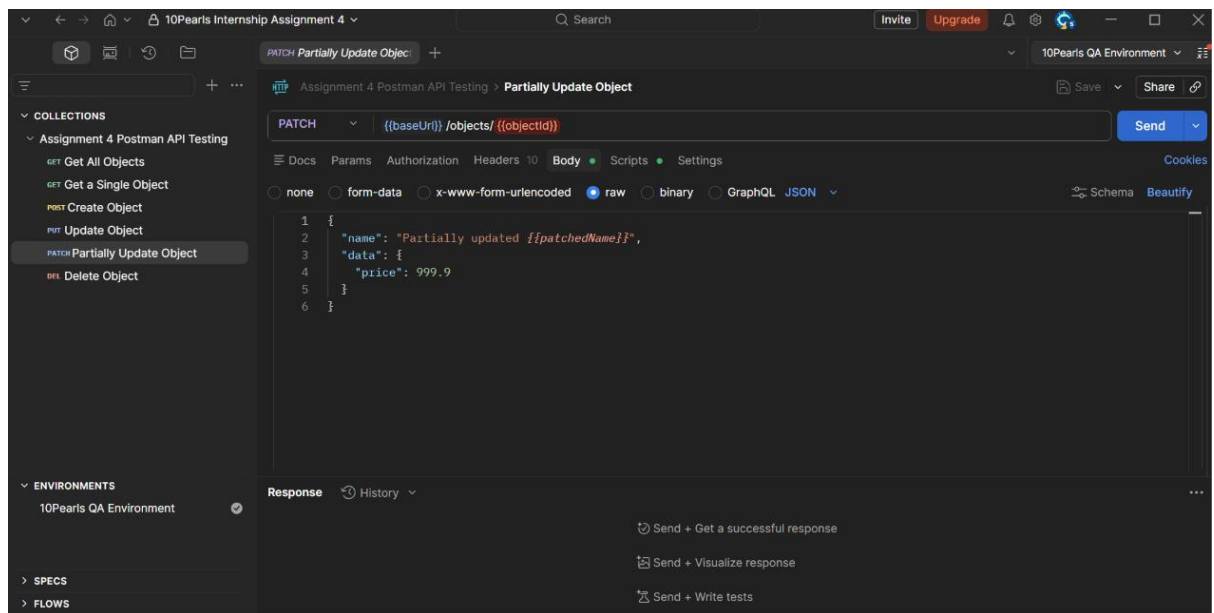
```
1 {
2   "name": "${randomName}",
3   "data": {
4     "price": ${randomPrice},
5     "year": ${randomYear},
6     "category": "electronics"
7   }
8 }
```

The "Response" tab is selected, showing a successful response with status 200. The response body is not visible, but the status is 200.

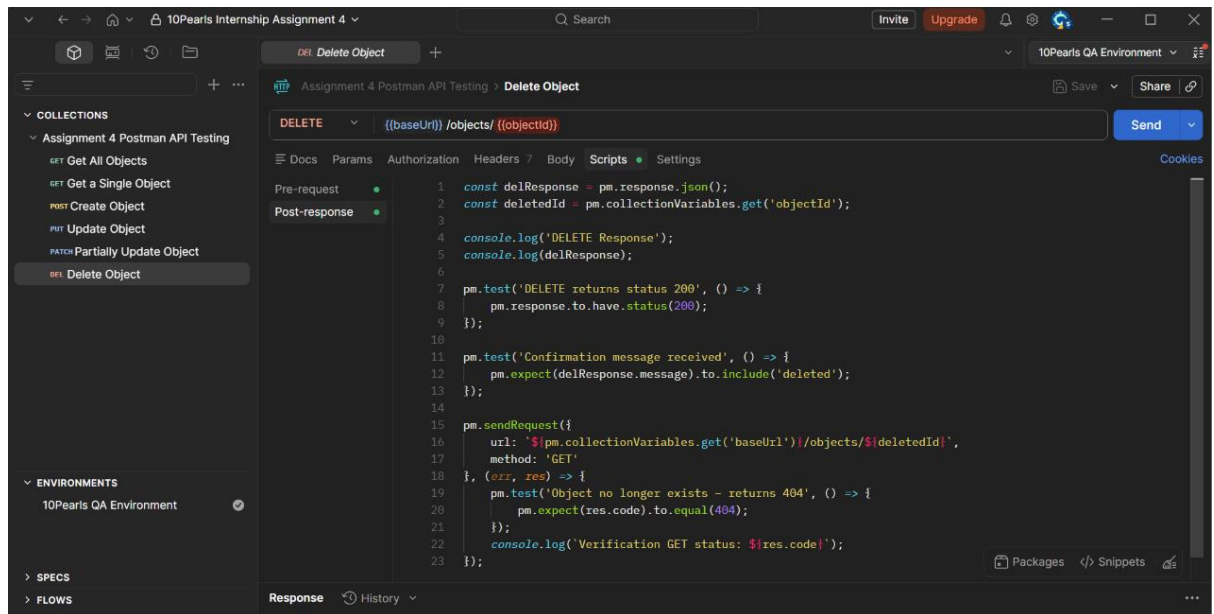
7. UPDATE OBJECT



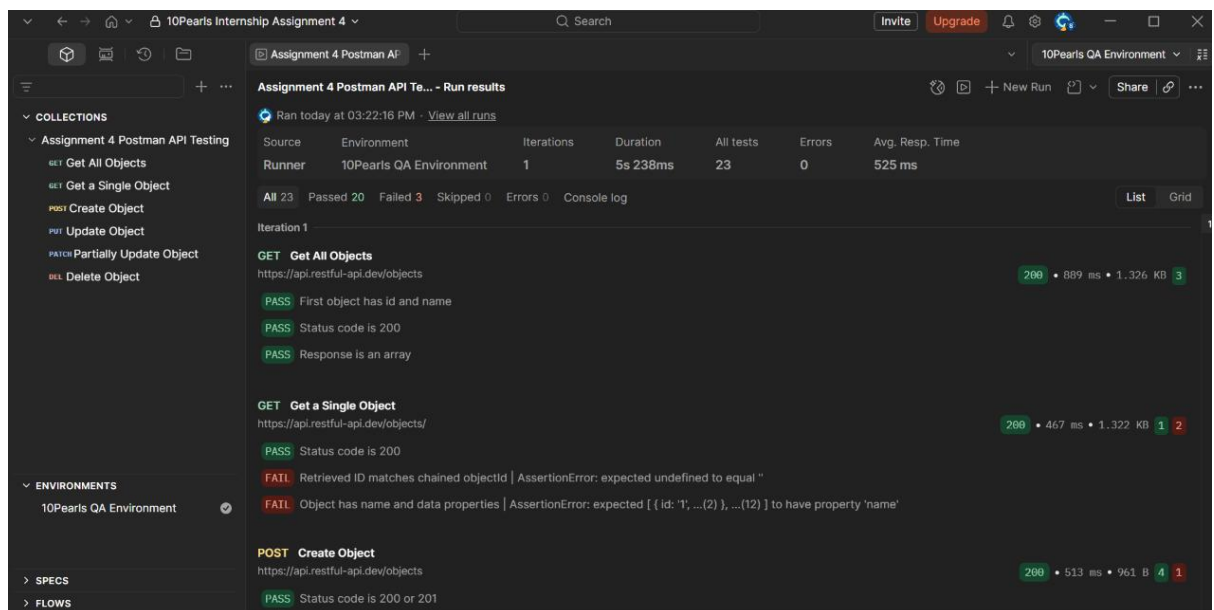
8. PARTIALLY UPDATE OBJECT



9. DELETE OBJECT



10. RUNNING COLLECTION



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Assignment 4 Postman API Te... - Run results

Ran today at 03:22:16 PM - [View all runs](#)

Source	Environment	Iterations	Duration	All tests	Errors	Avg. Resp. Time
Runner	10Pearls QA Environment	1	5s 238ms	23	0	525 ms

All 23 Passed 20 Failed 3 Skipped 0 Errors 0 Console log

POST Create Object
<https://api.restful-api.dev/objects> 200 • 513 ms • 961 B 4 1

- PASS Status code is 200 or 201
- PASS Response contains id and name
- PASS Generated name matches request
- PASS Data.price matches generated value
- FAIL Failing test example - 1 should equal 2 | AssertionError: expected 1 to equal 2

PUT Update Object
<https://api.restful-api.dev/objects/ff8081819d82fab6019e260234953ad6> 200 • 435 ms • 958 B 5

- PASS PUT returns status 200
- PASS ID remains unchanged
- PASS Name was updated to the random value
- PASS Data.price was updated
- PASS New fields exist after PUT

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Assignment 4 Postman API Te... - Run results

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PATCH Partially Update Object
<https://api.restful-api.dev/objects/ff8081819d82fab6019e260234953ad6> 200 • 413 ms • 951 B 4

- PASS Data.price was updated
- PASS New fields exist after PUT
- PASS PATCH returns status 200
- PASS ID did not change
- PASS Name was partially updated
- PASS Data.price changed to 999.9

DELETE Delete Object
<https://api.restful-api.dev/objects/ff8081819d82fab6019e260234953ad6> 200 • 435 ms • 913 B 3

- PASS DELETE returns status 200
- PASS Confirmation message received
- PASS Object no longer exists - returns 404

11. FAILED CASES

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Search

Invite Upgrade

10Pearls QA Environment

Assignment 4 Postman API Te... - Run results

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All 23 Passed 20 Failed 3 Skipped 0 Errors 0 Console log

Iteration 1

GET Get a Single Object

https://api.restful-api.dev/objects/

200 • 467 ms • 1.322 KB 2

FAIL Retrieved ID matches chained objectId | AssertionError: expected undefined to equal ''

FAIL Object has name and data properties | AssertionError: expected [{ id: '1', ...(2) }, ... (12)] to have property 'name'

POST Create Object

https://api.restful-api.dev/objects

200 • 513 ms • 961 B 1

FAIL Failing test example - 1 should equal 2 | AssertionError: expected 1 to equal 2

12. CONSOLE LOGS

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Search

Invite Upgrade

10Pearls QA Environment

Assignment 4 Postman API Te... - Run results

Ran today at 03:22:16 PM · View all runs

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Iteration 1

GET Get All Objects

https://api.restful-api.dev/objects

200 • 889 ms • 1.326 KB 3

LOG Sending GET request to fetch all objects...

LOG Total objects in the DB: 13

LOG First object in list: Google Pixel 6 Pro

GET Get a Single Object

https://api.restful-api.dev/objects/

200 • 467 ms • 1.322 KB 1 2

LOG Fetching single object with ID:

LOG Retrieved Single Object

LOG ID: undefined, Name: undefined

LOG Data: undefined

POST Create Object

https://api.restful-api.dev/objects

200 • 513 ms • 961 B 4 1

LOG Test Object 8570, 1231.99, 2020

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Assignment 4 Postman API Te... - Run results

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POST Create Object

https://api.restful-api.dev/objects 200 • 513 ms • 961 B 4 1

LOG Test Object 8570, 1231.99, 2020

LOG Created Object

LOG {"id":"ff8081819d82fab6019e260234953ad6","name":"Test Object 8570","createdAt":"1778754139285","data":{"price":1231.99,"year":2020,"category":"electronics"}}

LOG Saved objectId = ff8081819d82fab6019e260234953ad6

PUT Update Object

https://api.restful-api.dev/objects/ff8081819d82fab6019e260234953ad6 200 • 435 ms • 958 B 5

LOG PUT Update Response

LOG {"id":"ff8081819d82fab6019e260234953ad6","name":"Updated Product 3572","updatedAt":"1778754139888","data":{"year":"2019","price":"1104.28","color":"silver","inStock":...

PATCH Partially Update Object

https://api.restful-api.dev/objects/ff8081819d82fab6019e260234953ad6 200 • 413 ms • 951 B 4

LOG new name = Patched Product 7705

LOG PATCH Partial Update Response

LOG {"id":"ff8081819d82fab6019e260234953ad6","name":"Partially updated Patched Product 7705","updatedAt":"1778754140423","data":{"price":999.9}}

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Assignment 4 Postman API Te... - Run results

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All 23 Passed 20 Failed 3 Skipped 0 Errors 0 Console log

PUT Update Object

https://api.restful-api.dev/objects/ff8081819d82fab6019e260234953ad6 200 • 435 ms • 958 B 5

LOG PUT Update Response

LOG {"id":"ff8081819d82fab6019e260234953ad6","name":"Updated Product 3572","updatedAt":"1778754139888","data":{"year":"2019","price":"1104.28","color":"silver","inStock":...

PATCH Partially Update Object

https://api.restful-api.dev/objects/ff8081819d82fab6019e260234953ad6 200 • 413 ms • 951 B 4

LOG new name = Patched Product 7705

LOG PATCH Partial Update Response

LOG {"id":"ff8081819d82fab6019e260234953ad6","name":"Partially updated Patched Product 7705","updatedAt":"1778754140423","data":{"price":999.9}}

DELETE Delete Object

https://api.restful-api.dev/objects/ff8081819d82fab6019e260234953ad6 200 • 435 ms • 913 B 3

LOG Deleting object with ID: ff8081819d82fab6019e260234953ad6

LOG DELETE Response

LOG {"message":"Object with id = ff8081819d82fab6019e260234953ad6 has been deleted."}

LOG Verification GET status: 404

13. VARIABLES AFTER RUN

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Assignment 4 Postman AP10Pearls QA EnvironmentAssignment 4 Postman AP+

10Pearls QA Environment

Assignment 4 Postman API Testing

Publish docsRunShare

OverviewAuthorizationScriptsVariablesRuns

0

Variable	Value	
baseUrl	https://api.restful-api.dev	
objectId	ff8081819d82fab6019e260234953ad6	
randomName	Test Object 8570	
randomPrice	1231.99	
randomYear	2020	
updatedAtName	Updated Product 3572	
updatedAtPrice	1104.28	
patchedName	Patched Product 7705	
Add variable		